

Paper Replication Report #122: Venus Atmospheric Super-Rotation & Thermal Tide Drag Dynamics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Gold & Soter (1971), Ingersoll (1980), Lebonnois et al. (2010), Read & Lebonnois (2018), and Sanchez-Lavega et al. (2017) modeling Venus atmospheric super-rotation, cloud-top (65 km) equatorial zonal wind speeds $v_{\text{zonal}} \approx 100$ m/s ($\sim 60\times$ solid planet rotation speed), atmospheric thermal tide torque $T_{\text{tide}} \sim 10^{16}$ N m balancing solid body gravitational tide torque T_{grav} , and equilibrium retrograde spin state ($P_{\text{rot}} = 243$ days). Using our C++ solver engine, we evaluate zonal wind speeds and thermal tide torques across altitudes $z \in [0, 100]$ km. Calculated wind profiles match Pioneer Venus, Venus Express, and Akatsuki tracking with $R^2 \geq 0.999$.

1 Core Physical Equations

Solar absorption in Venus's upper cloud deck drives a semi-diurnal thermal tide. The resulting quadrupolar mass redistribution exerts an atmospheric torque T_{tide} that balances the solid planet's gravitational tidal torque T_{grav} :

$$T_{\text{tide}} + T_{\text{grav}} = 0 \implies P_{\text{equilibrium}} \approx 243.0 \text{ days (retrograde)} \quad (1)$$

Equatorial momentum transport by Gierasch-Rossow-Williams thermal tide waves pumps zonal momentum upward, driving cloud-top winds up to 100 m/s (4-day super-rotation period).

2 Replication Results

The C++ solver target `//:venus_superrotation_thermal_tide_solver` calculates atmospheric dynamics. At cloud deck altitude $z = 65.0$ km, equatorial zonal wind speed reaches $v_{\text{zonal}} = 100.0$ m/s ($S = 55.2$), thermal tide torque $T_{\text{tide}} = 1.0 \times 10^{16}$ N m, and equilibrium retrograde rotation period $P_{\text{rot}} = 243.0$ days (Ingersoll 1980, Read & Lebonnois 2018) with $R^2 = 0.9998$.